

Sriram Hariharan Neelakantan

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PROFESSIONAL EXPERIENCE

Techmentee, Sr. Data Scientist | Harrison, NJ

Jul 2025 - Present

- Cut average response time from 3 min to 45 sec by developing an LLM-based multi-agent intent framework (LangGraph + Step Functions) automating tier-1 FinTech chats like balance and KYC queries
- Expanded automation to nine workflows by building API-driven Action Agents (Lambda + Step Functions) for refund, payment retry, and card block, removing human review for low-risk cases
- Boosted resolution accuracy by 12% via a human-in-the-loop feedback loop retraining classifier agents and refining Bedrock prompt logic with dynamic evaluation metrics

L.B. Foster Salient Systems, Data Scientist II (AI Engineer) | Dublin, OH

Dec 2024 - Jun 2025

- Engineered retrieval corpus from KBs, SOPs, and incident logs, chunked with metadata (wheel, axle, circuit, temp) and embedded via Pinecone; auto-refreshed through Databricks to enable explainable diagnostics ($\geq 90\%$ doc hit-rate)
- Designed a practical RAG architecture around LLaMA-2 that paired a fast Pinecone retriever with the generator model instead of reinventing a custom architecture
- Fine-tuned LLaMA-2 with parameter-efficient training on domain failures and technician notes, improving reasoning clarity by $\sim 20\%$ and tracked through our experiment workflow in MLflow.
- Benchmarked the system with domain experts using retrieval hit-rate, hallucination rate, and clarity scoring, hitting ≥ 90 percent retrieval accuracy and strong interpretability gains over the base model
- Collaborated in deploying the fine-tuned model and built a LangGraph workflow that combined retrieval with sensor and repair data, delivering traceable diagnostics with evidence citations and a 90% clarity score in production

Norfolk Southern Corp, Machine Learning Intern | Atlanta, GA

Sep 2022 - Dec 2022

- Utilized K-Nearest Neighbors, PCA, and K-Means within an agile framework to cluster and enhance the spatial analysis of 90k + rail-road intersections, significantly advancing infrastructure planning efforts
- Streamlined Exploratory Data Analysis with Pandas, NumPy, and Matplotlib to pinpoint over 50 anomalies among 30,000+ data points, achieving a 40% boost in clustering accuracy and optimizing project deliverables
- Delivered a 20% reduction in operational costs and improved logistical performance by identifying inefficiencies in freight distributions using logistic regression and time series forecasting with Amazon Redshift
- Enhanced railway operations' reliability and increased logistical precision by 40% through the development of an anomaly detection system using Apache Spark & MLlib, identifying and correcting outliers in freight data

Accenture, Data Analyst II | India

Mar 2020 - Jul 2021

- Architected a data pipeline using supervised ML models and Apache Spark for insurance campaign response studies, enhancing conversion rates by 5.5% through customized campaigns
- Drove lapse and retention risk assessments using Python (gradient boosting) and customer segmentation with SQL, uncovering key drivers of early terminations that informed policy changes and reduced early-lapse rates by 15%
- Improved quarterly risk assessment accuracy by 12% by implementing a gradient boosting model that predicts policyholder default probabilities, using demographic & historical claim data to refine risk mitigation strategies
- Automated data cleaning and preprocessing workflows using Alteryx, streamlining ETL pipelines to enhance data quality and achieve a 25% reduction in manual processing time
- Conducted portfolio exposure analysis using Python for scenario modeling, SQL for data aggregation, & Tableau for risk visualization, informing pricing, design changes that improved profitability by 10% and reduced lapse risk by 12%

Accenture, Data Analyst I | India

Jan 2018 - Mar 2020

- Collaborated on a catalog management efficiency project using SQL, boosting productivity by 25%, orchestrated a quality framework with cross-functional teams, enhancing efficiency by 15%
- Developed ARIMA-based time series forecasting models on 250k+ longitudinal therapy utilization records, enabling early signal detection of seasonal demand fluctuations and improving patient access by 8% through optimized resource allocation

- Deduced key cardiovascular biomarkers in product performance assessments using Python and Tableau, leveraging AWS S3 and Databricks for data, improving early-risk detection accuracy by 14% across 12k+ patient records
- Built Python-based validation frameworks to cross-check trial records against regulatory and protocol standards, reducing compliance review time by 35% and improving quality assurance scores by 20%
- Manipulated and analyzed large-scale clinical and real-world datasets (trial records, biomarker studies) using Python, SQL, and Databricks, ensuring compliance with regulatory standards and improving early risk detection by 10%

ACADEMIC PROJECTS

Strategic Analysis of Hospitality Data | EDA, Classification, Time Series Analysis

- Employed SVM, Logistic Regression, and XGBoost to achieve high accuracy and performance in analyzing Expedia hotel data, complemented by in-depth EDA focusing on Average Daily Rate and cancellation trends using ARIMA

Healthcare Bot | LLM, CUDA

- Implemented HealthBot with FLAN-T5, GPT-2, LSTM, and BERT for enhanced intent recognition and entity detection, integrating user feedback to boost diagnostic accuracy and response

Topic Modeling - News Data | USML, Clustering

- Achieved an impressive coherence value of 0.68 across 14 well-defined topics by utilizing advanced techniques, including LDA, LSA, K-means clustering, and NMF alongside TF-IDF vectorization

SKILLS

Programming & Frameworks: Python (NumPy, Pandas, Scikit-Learn, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex), R, PySpark, Java (REST APIs), Shell Scripting, Git, Bitbucket, JIRA, Linux (UNIX)

LLMs & Generative AI: Large Language Models (GPT, LLaMA), Prompt Engineering, Fine-Tuning & PEFT, LoRA, Retrieval-Augmented Generation (RAG), Agentic AI Workflows, Multi-Agent Systems, OpenAI SDK, Hugging Face Hub, LangChain, LlamaIndex,, Gradio, Streamlit

Vector Databases & Embeddings: FAISS, Pinecone, OpenAI Embeddings, Cohere Embedding, Terraform

Data Engineering & Cloud Technologies: SQL, PostgreSQL, MySQL, MongoDB, Apache Spark, Databricks, Hadoop, AWS (S3, EC2, Lambda, Bedrock, Redshift), GCP (Vertex AI, BigQuery), ETL & Data Pipeline Orchestration, Automation Frameworks

Machine Learning & Analytics: Supervised / Unsupervised Learning, Deep Learning (PyTorch / TensorFlow), NLP (Text Classification, Summarization, Embeddings, NER), Predictive Modeling, Time Series Forecasting (ARIMA, Prophet), Statistical Testing, Causal Inference

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Plotly , Excel, RWE Dashboards

EDUCATION

Master of Science, MS in Data Science

Sep 2021 - Dec 2023

Northeastern University, Boston, MA